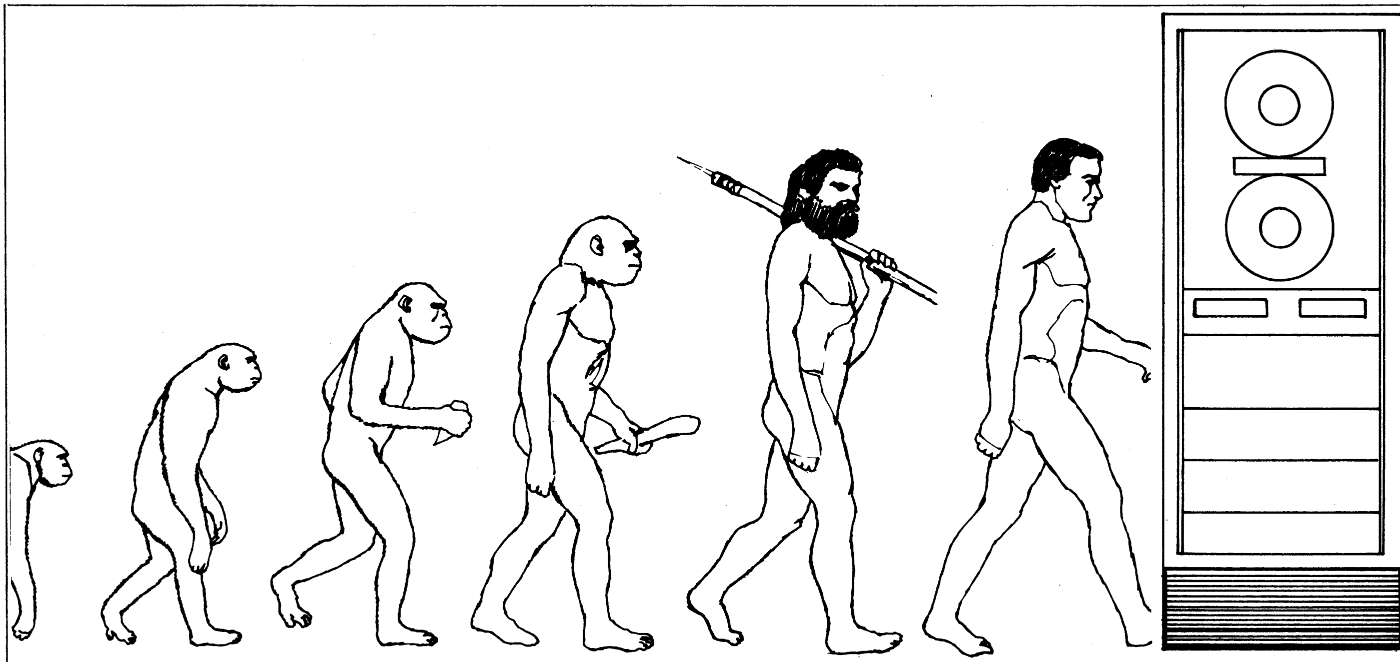
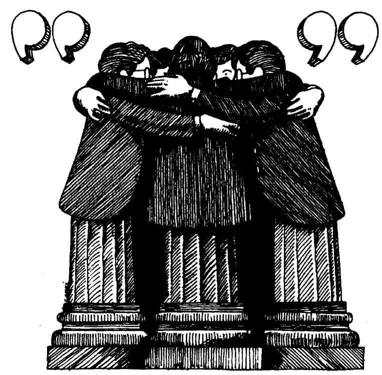


CATCH A RISING PROBLEM...
AND NEVER EVER LET IT GO



the forum on SOCIAL IMPLICATIONS

Harry T. Larson

A SERIOUS INDICTMENT

Two years ago I started a gut-level investigation of the *real* reasons senior computer engineers have chosen not to become involved in the matter of the social implications of computers. I focussed on two dozen 15-20-year veterans who I know well and with whom I have worked. I persisted until I got honest, pragmatic answers (even if it took until 2 A.M. and several drinks).

"There Are Smarter Ways To Advance My Career."

Most men start with the premise that they should select their professional activities so as to most rapidly advance their computer career. For most, this means focussing their professional and related energies on the work itself, their company, their working colleagues, their bosses, etc. For these people, work in professional societies makes no sense, let alone work on social implications.

Fair enough. For men with this sensible, normal, clear-cut set of priorities, it falls on deaf ears to suggest that computer engineers bear some responsibilities for the effects of our machinery on society, or enough responsibility that it calls for some investment of energy on this front.

"It's Too Controversial."

Its true that getting into the social implications arena plunges one into an extraordinary array of controversial matters. Social implications discussions very quickly broaden into bigger questions having to do with the general problems of our society, or problems that have been vexing mankind for thousands of years. Many engineers are passive about such matters. But most, though inactive, have *opinions* on such topics, and the opinions conflict with one another. Wham; the argument begins! Many such discussions have strong emotional content, sometimes fury.

So here we have the school of thought that says "avoid controversy — it can't help one's career, and the odds are it will hurt." Probably right. Very pragmatic. Play it safe. Stay as friendly as possible with as many people as possible. It's tough enough to achieve this in the normal course of life, and one certainly ought not to go out of his way to incur hostility.

The tough jobs almost always involve controversy, so it comes as an inescapable part of the work. However, given enough motivation, anyone can accept and deal with controversy. But of course therein lies the fundamental question here — motivation. While a few of the veterans are the play-it-safe-at-all-times types, most are highly motivated men, can-do types, willing and able to bull through resistance and controversy in their work. But in the social implications area, the motivation simply isn't there.

"It Will Get Me Crossways With My Management."

The general thread of this argument is that a truly objective pursuit of the effects of computers will sometimes force one to face and do something about unsavory or clearly bad uses of our systems, or bad effects of computers on people or society. If I take a position or take action, this will sometimes conflict with a sale of my company's systems, or otherwise create a conflict with the interests of my management.

As a professional and a member of society, I (HTL) don't buy this argument. But there it is, and its so real to many people that it stays their hand.

A more general facet of this viewpoint is interrelated with the controversy aspect. Some men feel, and other men know, that their management would prefer that they not become involved in potentially controversial activities, because it might reflect unfavorably on the company or organization.

"Harry, If I Were to Take Part in Computer Group Activities, I Sure as Hell Would Put My Effort In Places Where it Would Do My Career a Lot More Good."

For those who see value in making contacts and the value of technical exchange, work in local Chapters or on the Technical Committees makes more sense than working on Social Implications. For those who see value in the added dimension of national recognition, it makes more sense to work up the Computer Group hierarchy to some office in the national Group organization, or it makes more sense to work on a regional or national Conference.

The response above comes from men representing the 95% of computer engineers who have chosen not to be active in professional society work. But essentially the same answer is given by those who have for one reason or another become active, including those who do this work because of their belief that they are making a useful contribution to their profession.

For those who consider that professional society activity is inconsistent with being effective in their work, because it consumes time better spent on their job, I sometimes in desperation fall back on personal experience as valid (though unseemly and self-serving) evidence to the contrary. I point to the wide variety of successful R&D projects I've contributed to or directed, the variety of equipment I've developed, tons of systems in place and working, some bona fide pioneering in hardware and software, a variety of first-time and successful applications of computers and related technology, and steady managerial progress. And this has been done concurrent with 22 years of professional society activity.

But this argument is presented to no avail to those who have a mental set that professional society activity gets in the way of real productivity in their work.

To those who are active in professional society activities, I again occasionally fall back on personal experience as proof, that one can work in social implications *and* in other ways, pointing to my activity on technical committees, national chairmanship of the Computer Group, Chapter activities, editing of professional publications, chairing sessions, running technical programs, serving on the AFIPS Governing Board, etc. Thus, I can trot out at least personal proof that work on social implications is not mutually exclusive with other forms of technical society activity.

To no avail. Though boastful, its accurate, but this last ditch argument is not a winner.

What troubles me most about this is that these senior men are the men we need most, to be effective in the social implications area, for reasons given below.

"I Know One Thing About Human Nature — In Any Individual Everything Starts With No. 1."

This response continues to state open skepticism that anyone can really be interested enough in the problems of others to invest time in this endeavor. These men were cynical about my efforts, classified them as obviously phony, because the views I put forward are so obviously inconsistent with the fundamental "Big I — I come first."

For these people, anyone who works or shows interest in the social implications of computers is simply not for real. For such people, personal involvement in social implications is laughable.

Indictment

So I ferreted out some honest, bedrock, no-holds-barred answers.

For many of you, these probably make a lot of sense. I look upon them as a serious indictment of computer engineers and the organizations they work for.

Harry Kleinberg wrote in his letter to Brian Pollard (Sept.-Oct. Issue, COMPUTER), about the stereotype engineer, ". . . but the guy inside this pedestrian, glamorless shell is one of the true revolutionaries of history. He's a trouble-maker of epic proportions. He generates change. Not the trivial kinds of change that come and go as fads and styles, but deep, basic changes that affect the very way in which our society is structured . . . His computers can guide men to the moon and can reduce them to "frustrated," nameless digits. I could go on and on. In all these developments that are shaking our social order, the entrepreneur, labor and others play important parts, but the real root (radical) force is the engineer. In the cities and campuses the flamboyant characters in flowing hair and sandals aren't making a revolution — they're trying desperately to adjust to one. And the man who made it is the engineer."

As I have so often stated, with this goes some sort of fundamental responsibility. People are afraid of our machines, and some are becoming violent about it, and we have to make them unafraid. Some people are hurt by the coming of our machines, and we have some sort of responsibility to help ameliorate that hurt. Though our machines are neutral, and can be used for good or bad, just as any other machines, they are awesomely powerful in many respects, and hence can be more powerfully evil, in evil hands, creating an additional element of responsibility.

Some few engineers are interested enough to attend sessions on social implications. Some far fewer are interested enough to attend workshops. Some far fewer feel a sense of duty enough to serve on a panel, state a position, discuss it, defend it. Some far fewer will put their views in writing. (C'mon men, let's start sending those articles for this section of COMPUTER — I'm getting tired of writing them all.) And some far fewer will roll up their sleeves and work actively on a committee, make themselves available to give talks to PTA's and Scout Troops and Church groups and professional organizations, and get involved in meaningful ongoing time-consuming work in this area.

But all this is beginning to change. And the problems that call for change in engineers' attitudes are becoming more intense.

BASIS FOR REEXAMINATION

The younger engineers, like their peers in other walks of life, are showing real interest in the effects of computers on society. They have gratifying energy and enthusiasm, and we of course want to encourage that. But they need direction, and sometimes they actively seek it: "Tell me what I can do." This youthful enthusiasm often tends to produce impetuous actions, better left undone, or better done if moderated by an experienced hand. This youthful effort sometimes speaks with the omnipotence of youth, and serves us poorly; better it be strengthened with the wisdom of experience. This youthful activity sometimes is nothing more than an undirected crying out about what is wrong; better the same energy be channelled toward solutions, with the constructive goal-oriented guidance of successful engineers. This youthful activity is often impatient, and naively presumes that major changes can be achieved instantly; better that experience be added, giving broader knowledge of human nature and the fundamental aspects of society that pace change, leavening the impatience without stifling it. This youthful activity often runs out of gas if its first intense enthusiasm and drive does not meet with quick success; better it be taught to persevere; better its enthusiasm be aided to continue, by the supportive activity of older and toughened hands.

So we need more senior men involved, to make up the effective team on the social implications front. We need senior men for the reasons given above, and because you know "the system" and how to use it and combat it, and because you know the location of the handles that control the power of the establishment, and because senior men are in a better position to wield power.

Senior men are also needed for the reason that they are more bulletproof than junior engineers. More than once I've heard younger engineers express fear that they would incur the wrath of their superiors if they move forcefully into this controversial arena, and state that if some of their senior brethren would move openly and forcefully in the social effects of computers, it would make all the difference, make them (the younger engineers) feel safe, or strengthen their backbone enough that they would pitch in.

So for these reasons we need the leadership of more senior computer engineers.

Today some of the old reasons for not being involved are changing or disappearing. With the rapid onset of the nation's concern for our environment and the quality of life, activities that were formerly potentially controversial are now much more acceptable. With this, there is a much lesser risk of incurring the wrath of the organization one works for. Indeed, there are growing forces that make it in the best interest of organizations to become more active in a broader range of social activities. Forces afoot today in America make it more and more in the best interest of industry to put its best foot forward in community activities.

But more fundamental, I believe, is that more business leaders recognize that the energy, resources, genius, and can-do of business is one of the best hopes for mounting a really effective attack on now-critical social ills.

And because computers are becoming increasingly the target of extremists, because computers continue to create fear in people, and because computers represent to many people the most dangerous example of technology trampling on people, we are on the firing line, whether we like it or not. We must meet the destructive actions of the modern Luddities.

So I say to senior computer engineers, come on in, the water's fine, and the nation needs your help.

WHAT TO DO

The most constructive thing to do is to seek ways in which computers can help to solve age-old and modern problems. Pick a problem, and go to work on it, as an individual, or through your organizations. Consider my list of important problems as a starter:

Poverty: achieving more equitable distribution of our material goods.

Social welfare: breaking the vicious cycle that puts more and more people on social welfare rolls, and keeps them here, at taxpayer expense, and at the much greater expense of the destruction of the self esteem and lives of more and more people.

Medical practice: seeking to improve it, helping doctors to practice their profession more accurately, and helping to weed out those who practice it with deadly inaccuracy, against which our society presently has little or no recourse.

WORLD PEACE

Pollution: starting with present methods of computer monitoring, and moving on to employing computers in necessary studies yet to be done, and on to computer control where it can be helpful.

World population growth: aiding in the research needed to find ever more socially acceptable and widely applicable methods of ameliorating the problem identified by Malthus.

Representative government: helping to make our system ever more effective, as our nation grows, our society becomes more complex, and change accelerates.

TAXES

Soft "sciences": such as sociology and economics, where computers are needed to supply the missing link in true use of the scientific process, namely the test of hypothesis against reality, and thus make it possible for such sciences to move forward with the same sure tread as the hard sciences.

Ecology: aiding in furthering the research in the extrordinary complex multiple-looped cycles, which will surely call for the aid of our machinery.

Education: providing means for augmenting the teacher's work, personalizing education, and making it possible to take full advantage of the important times when a student is most interested, most receptive, most likely to be drawn onward.

DRUG CONTROL

Crime: aiding in the apprehension of criminals, and aiding in preventing crime.

Resources: seeking more efficient use of our dwindling natural resources, both in first use and "waste" use.

I suggest that you seize on a problem that interests you intensely (to say the obvious), perhaps one in which you or yours have been seriously harmed, providing motivation that will sustain you as you work on some facet of these gigantic problems. For those of you of the bent of mind that says these problems are too big for any one man, I suggest that you can always find a small enough piece of the problem, one on which you can make a detectable dent, or one which can be rectified in your lifetime, and thus plant ideas, and provide proof that will cause the ideas to be practical and spreadable.

Do more than breast beating — we have plenty of that already. Do more than doom-saying — we're already plentifully supplied with that too. Resolve to follow through to attain a solution to the problem you choose.

In this enterprise, seek to associate with reasonable men. Don't waste your time with zealots, nor with those whose only purpose is to propagandize. As Tom Patterson of RCA has pointed out, in the main, progress is made by reasonable men working with reasonable men, toward a goal.

Your actions can range from personal, one-man actions, to moving into the corridors of power (and we need more engineers there).

You might try Jim Brownlow's suggestion of assembling small groups of similarly interested professionals, in your organization or in your community. Set goals, then learn what you must learn. As engineers, we tend to be very narrow in our knowledge of society and people. We tend to be too quick to pronounce solutions in areas where we are truly ignorant. Take your time. Immerse yourself. Understand. *Then* start to solve, only then. These small groups might be interlinked with other such groups, and grow.

Another way is to persuade organizations to which you belong to work on the problem you choose, bringing your knowledge of computers to the table, seeking means to put it to work.

Seek continuing interdisciplinary meetings between your professional society and the professionals in the field wherein your target problem lies. Then spin off working groups to attack and solve the problem.

Write a paper, give a paper, publish a paper. Then latch onto those who show interest, as the nucleus of a problem-solving activity.

You might become alert to evil uses of computers, and undertake ways and means within the system to rectify such matters.

Work to become a member or advisor to a local planning commission, school board, or other appropriate agency, as a channel to tap the power structure for furthering your problem-solving enterprise.

If you feel properly prepared to reach higher into the corridors of power, make your computer expertise known to your local, state, or federal representatives, who are making decisions involving computers while they know little about them. Experience to date indicates that they will welcome the advice and counsel of experts in computerdom.

So much for a sampler of ways of taking individual action, or action via organizations.

When you think about it, the actions I'm suggesting here are simply an extension of the basic purpose of engineers — to apply the discoveries of science for the good of society. I'm urging that computer engineers apply *themselves* as well as their machines to this end.

The thing you'll need most is perseverance. The study and discourse and immersion in the problem area may take many years. Don't be scared off, offended, confused, or exasperated by the complications you encounter, as have engineers who have ventured forth before you. Don't lose sight of your goal, and do what engineers do well, working toward the goal until you have a solution. Fight for implementation of your solution .

Persevere.